Practical No. 01

ZOO 1203- General Entomology

Identification of Insect Orders

Objectives:

- Identifying insects by using their morphological features to order level

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1. Identify the order of the given insects by using the provided key.

2. Draw, label and write the classification of each specimen. Write the features that help you to identify the order of each specimen as the footnote.

1. Fly
2. Mosquito
3. Earwing
4. Beetle
5. Weevil
6. Shield Bug
7. Stink bug
8. Aphids
9. Cicadas
10. Butterfly
11. Moth
12. Thrips
13. Dragonfly
14. Damselfly
15. Mayfly
16. Stone fly
17. Termite
18. Caddisfly

KEY TO THE ORDERS OF ADULT INSECTS

1 Winged insects (wings sometimes hard and opaque covering some or all of the abdomen 2

Wingless insects (all intersegmental abdominal sutures uncovered) 18

2 (1) With only one pair of wings and these are membranous .... DIPTERA .... (two winged flies)

With two pairs of wings, or fore wings not membranous 3

3 (2) Fore wings hard and horny in texture, meeting in a straight line without overlapping and usually concealing membranous hind wings (certain forms have the hind wings vestigial or absent) 4

Fore wings not as above 5

4 (3) Abdomen provided with forceps-like cerci at posterior end .... DERMAPTERA .... (earwigs)

Abdomen without forceps-like cerci .... COLEOPTERA .... (beetles, weevils)

5 (3) Fore wings and hind wings different in texture and structure, forewing horny or leathery, hind wing membranous 6

Fore wings and hind wings usually similar in texture and structure, if forewing more leathery than hind wings then with piercing-sucking mouthparts 7

6 (5) Fore wings thick and leathery at base, without veins; membranous at tip. Mouthparts forming a sucking beak .... HEMIPTERA .... suborder Heteroptera (bugs)

Fore wings leathery and with veins throughout, with chewing or sucking mouthparts 26

7 (5) Wings partially, or more often entirely covered with microscopic overlapping scales. Mouthparts usually in the form of a coiled proboscis .... LEPIDOPTERA .... (butterflies, moths)

Wings largely transparent or covered with fine hair. Mouthparts not as above 8

8 (7) Wings very narrow, strap-like, fringed with long hairs, small slender bodied insects with cone-like mouthparts .... THYSANOPTERA
.... (thrips)

Wings and mouthparts not as above 9

9 (8) Mouthparts a piercing-sucking beak on bottom of the head, often resting on underside of body between the legs(Fig 1) .... HEMIPTERA
.... suborder Homoptera (aphids, cicadas etc)

Mouthparts **not** a piercing-sucking beak and normally situated at the front of the head 10

10 (9) Antennae small and bristle-like.. 11

Antennae conspicuous and of different forms.. 12

11 (10) Fore wing and hind wing nearly equal in size; tip of abdomen without terminal filaments .... ODONATA
.... (dragonflies, damsel flies)

Fore wings much larger than hind wings; tip of abdomen with 2-3 long terminal filaments .... EPHEMEROPTERA
.... (mayflies)

12 (10) Wings with many veins and cross veins in a net like arrangement (Fig 2) 13

Wings with few veins and cross veins with larger cells (clear areas between veins).. 17

13 (12) Hind tarsi with fewer than five segments.. 14

Hind tarsi with five segments.. 15

14 (13) Tarsi three-segmented, hind wing usually distinctly wider than fore wing, cerci many segmented.. PLECOPTERA
.... (stoneflies)

Tarsi four-segmented, fore wings and hind wings of equal size, cerci minute, with less than 9 segments .... ISOPTERA
.... (termites)

15 (13) Wings covered with fine hair, lacking margin of densely divided veins .... TRICHOPTERA
.... (caddisflies)

Wings usually transparent, not covered with hair, at most with scattered hairs mainly on forewing veins.. 16

16 (15) Wings lacking margin of densely divided veins. Hind wing considerably wider than forewing of base .... MEGALOPTERA
.... (dobsonflies)

Wings with a dense margin of bifurcating veins. Hind wing .... NEUROPTERA
usually narrower, sometimes as wide as forewing at base .... (lacewings)

17 (12) Tarsi two- or three-segmented, wings approximately equal in size, abdomen broadly joined to thorax .... PSOCOPTERA
.... (barklice)

Tarsi usually five-segmented, fore wings larger than hind wings, abdomen usually constricted at base .... HYMENOPTERA
.... (ants, bees, wasps)

18 (1) Abdomen composed of six or fewer segments with a ventral springing apparatus .... COLLEMBOLA
.... (springtails)

Abdomen with more than six segments 19

19 (18) Abdomen with three long terminal filamentous appendages; abdominal segments 2-7 may each have a pair of small ventral leg-like appendages .... THYSANURA
.... (Bristletails)

Abdomen with no more than two terminal appendages; segments 2-7 without ventral appendages 20

20 (19) Mouthparts of the chewing and biting type 21

Mouthparts of the piercing, lapping or sucking type, sometimes concealed 29

21 (20) Abdomen constricted at base .... HYMENOPTERA
.... (ants, wasps)

Abdomen not constricted at base but broadly jointed to thorax 22

22 (21) Louse-like insects (Fig 3), usually less than 6 mm in length 23

Insects not louse-like, usually larger 24

23 (22) Antennae with five or less segments, strongly flattened insects .... MALLOPHAGA
.... (chewing lice)

Antennae with more than five segments, not strongly flattened insects .... PSOCOPTERA
.... (barklice)

24 (22)Body superficially ant- like in form pale, whitish in colour and weakly sclerotized, part from head capsule....................ISOPTERA
.......................(termites)

Body shape various but not ant like (as above), usually well pigmented and sclerotized 25

25 (24)Abdomen with a pair of forceps – like cerci at posterior end....................DERMAPTERA
.......................(earwigs)

Abdomen without forceps – like cerci 26

26(6,25)Hind legs usually greatly enlarged for jumping....................ORTHOPTERA
.......................(rickets, wetas, grasshoppers etc)

Hind legs not so enlarged 27

27(26) Insects broadly dorsoventrally flattened, all legs adapted for running....................BLATTODEA
.......................(cockroaches)

Insects not dorsoventally flattened but markedly elongated,usually more than three times as long as wide 28

28 (27)Fore legs raptorial (specialized for grasping prey) prothorax usually narrow and elongate ....MANTODEA
.......................(praying mantids)

Fore legs not specialized as above, prothorax shorter than other thoracic segments ....PHASMATODEA
.......................(stick insects)

29 (20)Tarsi with five segments....................30
Tarsi with fewer than five segments 31

30(29)Body strongly compressed laterally ....................SIPHONAPTERA
.......................(fleas)

Body oftenly dorsoventrally flattened ....DIPTERA
.......................(Sheep ked and other parasitic flies)

31 (29) Last tarsal segment a bladder – like organ without distinct claws ....................THYSANOPTERA
.......................(thrips)

Last tarsal segment not bladder like with one or two claws....................32

32 (31) Louse-like form, mouthparts withdrawn into head .... ANOPLURA
.... (sucking lice)

Body not usually louse-like in form, mouthparts
in the form of a segmented beak-like proboscis .... HEMIPTERA
.... (bugs)